

• Easy # 26 → Remove Duplicates From Sorted Arrays

▲ Problem Statement → Given a sorted array, remove duplicates in-place so that each unique element appears only once.

Return the number of unique elements.

▲ Main Solution • Use two pointers:- • "i" checks every element.

• "j" keeps track of the last unique element.

When "nums[i]" is different from "nums[j]", move "j" forward & copy the new unique number there.

▲ Brute Force • Use another array to store only unique elements, then copy them back into "nums".

▲ Algorithm • 1. Set "j = 0"

2. Loop "i" from "1" to the end.

3. If "nums[i] != nums[j]" :- increment "j"

• Set "nums[j] = nums[i]"

4. Return "j + 1"

▲ Important Points • The array is already sorted, so duplicates are next to each other.

• No extra array is needed.

• "i" checks elements

• "j" stores the position of unique elements.

• Return "j + 1" because indexing starts from "0".

▲ Complexity → Brute Force → $O(n)$ & $O(n)$
 Main Solution → $O(n)$ & $O(1)$